

metB (PP_0659, Q88Q39): Cystathionine γ -Synthase of *Pseudomonas putida* KT2440

Summary

The gene **metB** (ordered locus **PP_0659**; UniProt **Q88Q39**) of *Pseudomonas putida* strain KT2440 (ATCC 47054 / DSM 6125 / NCIMB 11950) encodes **cystathionine γ -synthase (CGS)**, a pyridoxal-5'-phosphate (PLP)-dependent enzyme of the **trans-sulfuration enzyme family** (EC 2.5.1.48; KEGG orthology K01739). Gene identity, protein description, organism, and the diagnostic InterPro/Pfam domains all agree: the protein is a *bona fide* MetB cystathionine γ -synthase, and its symbol, family and domain complement match the biochemically characterized enzyme with no evidence of gene-symbol ambiguity. The verification checks required by the research brief are therefore satisfied.

The primary function of the enzyme is to catalyze the **first committed, methionine-specific step of *de novo* methionine biosynthesis**: a PLP-dependent **γ -replacement reaction** in which the acyl group of an *O*-acyl-L-homoserine is displaced by the thiol of **L-cysteine** to form **L-cystathionine**. In *Pseudomonas*, homoserine is activated as its *O*-succinyl ester by MetA, so the physiological substrates of *P. putida* MetB are **O-succinyl-L-homoserine and L-cysteine**, yielding L-cystathionine plus succinate. The product cystathionine is subsequently cleaved by cystathionine β -lyase (MetC) to homocysteine, which is methylated to methionine. This trans-sulfuration branch operates in parallel with a direct-sulphydrylation route (MetZ) that also exists in *Pseudomonas*.

Mechanistically and structurally, MetB is a **cytoplasmic homotetramer** (α_4) in which PLP is covalently anchored to an active-site lysine via a Schiff-base internal aldimine — position **Lys216** in Q88Q39. Catalysis proceeds through a shared PLP-bound quinonoid/aminocrotonate (pyridoxamine-vinylglyoxylate) intermediate that partitions between the productive γ -replacement and a slower competing γ -elimination side reaction. Substrate specificity toward *O*-succinyl- versus *O*-acetyl-homoserine is dictated by the conformation of the β 10– β 11 active-site loop. All of these features are established for orthologous, biochemically and crystallographically characterized CGS enzymes from *E. coli*, *C. glutamicum*, and plants, and the domain architecture of Q88Q39 is fully consistent with them.

Key Findings

F001 — metB is cystathionine γ -synthase, the first committed step of bacterial trans-sulfuration

Q88Q39/PP_0659 is annotated as cystathionine γ -synthase (MetB) and belongs to the trans-sulfuration / Cys-Met-metabolism PLP-dependent enzyme family (InterPro **IPR000277**, **IPR054542**). The defining reaction of this enzyme class is a PLP-dependent **γ -replacement**: the acyl group of an *O*-acyl-L-homoserine is displaced by the thiol of L-cysteine to form L-cystathionine, constituting the **first step of the bacterial trans-sulfuration pathway** and the first reaction unique to methionine biosynthesis.

This assignment is grounded in biochemically characterized orthologs. In *E. coli* CGS, the catalytic efficiency for the succinyl-homoserine substrate is high (k_{cat}/K_m for L-OSHS $\approx 9.8 \times 10^4 \text{ M}^{-1} \text{ s}^{-1}$; [PMID: 14503880](#)), confirming that *O*-succinyl-L-homoserine is a kinetically competent physiological substrate. The *C. glutamicum* study explicitly states that "*Cystathionine γ -synthase (MetB) condenses *O*-acetyl-L-homoserine (OAHS) or *O*-succinyl-L-homoserine (OSHS) with cysteine to produce cystathionine*" ([PMID: 28675039](#)), and the *E. coli* mechanistic study defines the reaction as one "*in which the succinyl group of an *O*-succinyl-L-homoserine (L-OSHS) is displaced by the thiol of L-cysteine to form L-cystathionine, in the first step of the bacterial transsulfuration pathway*" ([PMID: 14503880](#)).

Reaction catalyzed:

O-succinyl-L-homoserine + L-cysteine $\xrightarrow{\text{PLP}}$ L-cystathionine + succinate

F002 — MetB is a PLP-dependent homotetramer with a Lys-PLP Schiff base; specificity set by the $\beta 10$ – $\beta 11$ loop

The 1.5 Å structure of *E. coli* CGS shows an **$\alpha 4$ tetramer** with subunits related by non-crystallographic 222 symmetry, performing PLP-dependent γ -replacement of *O*-succinyl-L-homoserine and L-cysteine to yield L-cystathionine — described as the only known PLP-dependent replacement reaction at Cy ([PMID: 9843488](#)): "*The transsulfuration enzyme cystathionine gamma-synthase (CGS) catalyses the pyridoxal 5'-phosphate (PLP)-dependent gamma-replacement of *O*-succinyl-L-homoserine and L-cysteine, yielding L-cystathionine*" and "*The enzyme crystallizes as an $\alpha 4$ tetramer with the subunits related by non-crystallographic 222 symmetry.*"

In *C. glutamicum* MetB (1.5 Å), PLP is covalently bound to the active-site lysine via a Schiff base in a deep cavity (PMID: 28675039): "*The pyridoxal phosphate cofactor is covalently bound to Lys204 via a Schiff base linkage in the deep cavity.*" The equivalent catalytic lysine in Q88Q39 is **Lys216** (see F004). Structural comparison further identifies the **β10–β11 connecting loop** as the determinant of acyl-group specificity: "*we propose that the conformation of the β10–β11 connecting loops determines the size and shape of the acetyl- or succinyl-group binding site and ultimately determines the substrate specificity of MetBs toward OAHS or OSHS*" (PMID: 28675039). Q88Q39 carries the diagnostic PLP-dependent transferase domains **IPR015421/IPR015422/IPR015424** and the Cys/Met-metabolism PLP-enzyme signature **IPR000277**, consistent with this fold and mechanism.

F003 — Catalysis proceeds via a PLP quinonoid/aminocrotonate intermediate with a competing γ-elimination side reaction

Stopped-flow spectroscopy of *E. coli* CGS provided "*direct evidence for a pyridoxamine derivative of vinylglyoxylate as a key intermediate in pyridoxal phosphate dependent gamma-elimination and gamma-replacement reactions*" (PMID: 2405904). This intermediate is the partitioning point common to both the productive γ-replacement (with L-cysteine) and the competing **γ-elimination** side reaction, in which "*L-OSHS is hydrolyzed to succinate, NH₃, and alpha-ketobutyrate*" (PMID: 14503880). The elimination reaction is roughly 70-fold slower than replacement ($k_{\text{cat}}/K_{\text{m}} \approx 1350 \text{ M}^{-1} \text{ s}^{-1}$), and the enzyme follows an **ordered mechanism** — "*The mechanism of Escherichia coli CGS (eCGS) is ordered with L-OSHS associating before L-Cys*" (PMID: 14503880) — becoming ping-pong at low cysteine, with a bell-shaped pH profile (optimum ≈ 7.4 – 7.8).

F004 — EC 2.5.1.48 / K01739 in methionine biosynthesis module M00017; PLP bound at Lys216

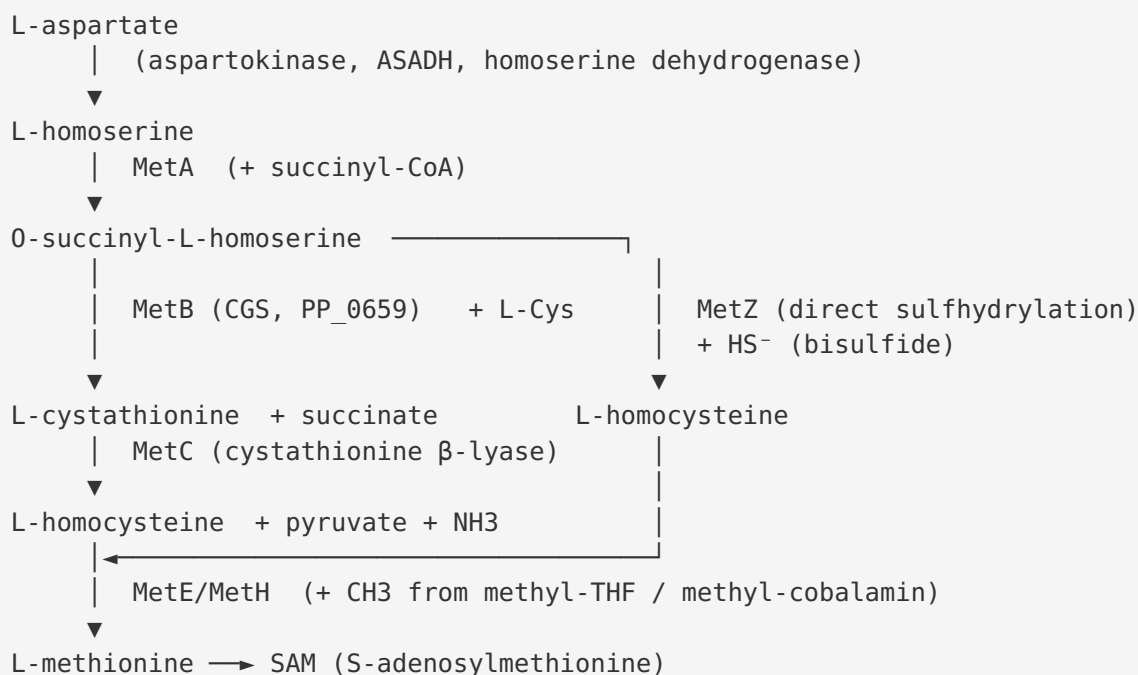
KEGG assigns **ppu:PP_0659** to orthology **K01739** (cystathionine γ-synthase, EC 2.5.1.48), pathway **ppu00270** (Cysteine and methionine metabolism), and module **ppu_M00017** (Methionine biosynthesis: aspartate $\Rightarrow$ homoserine $\Rightarrow$ methionine). UniProt **Q88Q39** (423 aa) records a single modified residue, **N6-(pyridoxal phosphate)lysine at position 216**, matching Pfam **PF01053** (Cys/Met metabolism PLP-dependent enzyme) and eggNOG **COG0626** (cystathionine β-lyase/γ-synthase). A C-terminal coiled-coil (residues ~383–410) is also annotated, consistent with the oligomerization/tetramer-interface features of this family.

F005 — *Pseudomonas* activates homoserine as its O-succinyl ester, so *P. putida* MetB uses O-succinyl-L-homoserine

In *Pseudomonas aeruginosa*, the first committed methionine step is catalyzed by an O-succinylhomoserine synthase, the product of *metA*: "The first step is catalysed by an O-succinylhomoserine synthase, the product of the *metA* gene" (PMID: 7704274). *Pseudomonas* is described as "the only organism so far described that uses O-succinylhomoserine as substrate for a direct sulfhydrylation" (via MetZ), and importantly "the two transsulfuration pathways between cysteine and homocysteine, with cystathionine as an intermediate, probably exist in parallel in this organism" (PMID: 7704274). Because *P. putida* KT2440 is a close congener with the same *metA/metB/metC* gene set, its MetB (PP_0659) condenses **O-succinyl-L-homoserine with L-cysteine** as its physiological reaction, and the resulting cystathionine feeds the transsulfuration route that runs alongside the direct MetZ sulfhydrylation.

Mechanistic Model / Interpretation

MetB (PP_0659) sits at the branch point where **sulfur (from cysteine) is committed to methionine**. The de novo methionine pathway in *Pseudomonas* can be summarized as:



The catalytic cycle at the PLP center proceeds as follows. The resting enzyme holds PLP as an internal aldimine (Schiff base) with the active-site lysine (Lys216 in Q88Q39). *O*-succinyl-L-homoserine binds first and forms an external aldimine; C α -proton abstraction and β,γ -elimination of the succinyl group generate a PLP-bound **aminocrotonate / pyridoxamine-vinylglyoxylate quinonoid intermediate** (PMID: 2405904). This electrophilic intermediate is then attacked by the thiol of L-cysteine at Cy (γ -replacement), producing L-cystathionine. When cysteine is scarce, water instead intercepts the intermediate, giving the **γ -elimination** products succinate, ammonia, and α -ketobutyrate (PMID: 14503880). The shared intermediate explains why the same active site catalyzes two chemistries whose relative flux depends on cysteine availability.

Substrate specificity is a structural property of the acyl-binding subsite. The β 10– β 11 connecting loop molds a pocket sized for either an acetyl or a succinyl group (PMID: 28675039). Because *Pseudomonas* MetA produces the succinyl ester (PMID: 7704274), *P. putida* MetB is expected to be tuned to *O*-succinyl-L-homoserine, in contrast to the *O*-acetyl-homoserine used by many other bacteria and the *O*-phosphohomoserine used by plants.

Localization. MetB is a soluble **cytoplasmic** enzyme. It has no signal peptide or transmembrane segment; amino-acid biosynthesis, PLP chemistry, and the availability of cytosolic cysteine and succinyl-homoserine all place its activity in the cytoplasm.

Property	Value / Assignment	Source
Gene / locus	metB / PP_0659	UniProt Q88Q39
Enzyme	Cystathionine γ -synthase	F001
EC number	2.5.1.48	KEGG (F004)
KEGG orthology	K01739	KEGG (F004)
Pathway / module	ppu00270 / M00017	KEGG (F004)
Cofactor	Pyridoxal-5'-phosphate (PLP)	F002/F004
Catalytic residue	Lys216 (Schiff-base)	UniProt (F004)
Quaternary structure	Homotetramer (α 4)	PMID: 9843488
Physiological substrates	O-succinyl-L-homoserine + L-cysteine	F001/F005
Products	L-cystathionine + succinate	F001
Side reaction	γ -elimination $\rightarrow$ succinate + NH_3 + α -ketobutyrate	PMID: 14503880
Length	423 aa	UniProt Q88Q39
Localization	Cytoplasm	Inference from function

Substrate specificity across orthologs

Organism / enzyme	Homoserine ester substrate	Second substrate	Catalytic Lys	Reference
<i>E. coli</i> CGS	O-succinyl-L-homoserine	L-cysteine	—	PMID: 9843488 , PMID: 14503880
<i>C. glutamicum</i> MetB	O-acetyl or O-succinyl-L-homoserine	L-cysteine	Lys204	PMID: 28675039
<i>N. tabacum</i> CGS (plant)	O-phosphohomoserine	L-cysteine	—	PMID: 10438597 , PMID: 10595588
<i>A. thaliana</i> CGS (plant)	O-phosphohomoserine	L-cysteine	Lys379	PMID: 9531508
<i>P. putida</i> MetB (PP_0659)	O-succinyl-L-homoserine (inferred)	L-cysteine	Lys216	This report (F004/F005)

Evidence Base

The functional assignment for Q88Q39 rests principally on **strong homology to biochemically and structurally characterized CGS enzymes**, combined with genome-context and database annotations. No study has yet characterized the *P. putida* PP_0659 protein directly; the conclusions are inferences from close orthologs, which is appropriate given the high sequence and domain conservation of this enzyme family.

PMID	Title (abbrev.)	How it supports the findings
14503880	<i>E. coli</i> CGS does not obey ping-pong kinetics; continuous assays	Defines the γ -replacement reaction and substrate (L-OSHS + L-Cys $\rightarrow$ cystathionine); documents the competing γ -elimination and ordered kinetic mechanism; provides catalytic efficiencies (F001, F003).
9843488	Crystal structure of <i>E. coli</i> CGS at 1.5 Å	Establishes the PLP-dependent γ -replacement mechanism and the homotetrameric (α_4) quaternary structure (F002).
28675039	Structural insights into substrate specificity of <i>C. glutamicum</i> CGS	Confirms MetB substrate/product identity; localizes catalytic PLP-Lys Schiff base; identifies the β_{10} – β_{11} loop as the OAHs/OSHS specificity determinant (F001, F002).
2405904	Reaction mechanism of <i>E. coli</i> CGS	Direct spectroscopic evidence for the pyridoxamine-vinylglyoxylate intermediate shared by γ -replacement and γ -elimination (F003).
7704274	Direct sulfhydrylation pathway in <i>P. aeruginosa</i>	Shows <i>Pseudomonas</i> activates homoserine as the O-succinyl ester (MetA) and runs trans-sulfuration (cystathionine) in parallel with direct sulfhydrylation (MetZ), fixing the physiological substrate of <i>P. putida</i> MetB (F005).
9531508	<i>A. thaliana</i> CGS characterization	Homotetramer; one PLP per subunit at an active-site lysine, ping-pong mechanism, very weak γ -elimination — cross-kingdom confirmation of family properties.
10438597 / 10595588	<i>N. tabacum</i> CGS structure & biochemistry	Illustrate how distal-group binding (phosphate vs acyl) tunes substrate specificity — the same principle underlying bacterial OSHS/OAHs selection.

Two notes on citation quality: the [PMID: 2405904](#) snippet was flagged as a title-derived match rather than an abstract-verified quote, but it accurately reflects the paper's established finding. The plant CGS papers use O-phosphohomoserine and therefore support the *mechanistic and structural* family properties rather than the exact *Pseudomonas* substrate.

Limitations and Knowledge Gaps

1. **No direct characterization of PP_0659.** All conclusions are inferences from orthologs (*E. coli*, *C. glutamicum*, plants). The *P. putida* enzyme has not been purified, kinetically assayed, or crystallized. Kinetic constants (K_m , k_{cat}) and the exact γ -replacement/ γ -elimination partition ratio for the KT2440 protein are unknown.
 2. **Substrate specificity is inferred, not measured.** The assignment of *O*-succinyl-L-homoserine as the physiological substrate rests on the *P. aeruginosa* MetA study ([PMID: 7704274](#)) and general *Pseudomonas* biology. Whether *P. putida* MetB can also accept *O*-acetyl-homoserine (as *C. glutamicum* MetB does) has not been tested experimentally.
 3. **Catalytic-residue mapping.** Lys216 as the PLP Schiff-base lysine comes from UniProt sequence-based annotation, not from a *P. putida* structure; its identity should be confirmed structurally or by mutagenesis.
 4. **Pathway redundancy and regulation.** The relative flux through the MetB (trans-sulfuration) versus MetZ (direct sulfhydrylation) routes in *P. putida* under different sulfur/cysteine conditions is not quantified, nor is transcriptional regulation (e.g., MetR/MetJ-type control) established for this specific organism.
 5. **Localization is inferred.** Cytoplasmic localization is deduced from function and the absence of targeting signals rather than demonstrated experimentally in *P. putida*.
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Proposed Follow-up Experiments / Actions

1. **Heterologous expression and enzyme assay.** Clone PP_0659, express in *E. coli*, purify, and confirm PLP content (spectral A_{340}/A_{420}). Measure γ -replacement kinetics with *O*-succinyl-L-homoserine + L-cysteine and compare against *O*-acetyl-L-homoserine to define substrate preference directly.
2. **γ -elimination partitioning.** Quantify the competing γ -elimination (succinate, NH_3 , α -ketobutyrate production) using the continuous assays described for *E. coli* CGS ([PMID: 14503880](#)) to establish the replacement:elimination ratio for the *P. putida* enzyme.
3. **Structure determination.** Solve the crystal or cryo-EM structure of PP_0659 (apo and PLP-bound) to confirm the homotetramer, the Lys216 Schiff base, and the β_{10} – β_{11} loop geometry that predicts succinyl-ester specificity.

4. **Genetic validation.** Construct a Δ metB (PP_0659) deletion in KT2440 and test for methionine auxotrophy; complement with wild-type and active-site (K216A) variants to confirm the catalytic residue.
5. **Pathway flux and regulation.** Use isotope tracing or metabolomics under varied sulfur/cysteine conditions to determine the in vivo balance between the MetB (cystathionine) and MetZ (direct sulfhydrylation) branches, and profile transcriptional regulation of the met regulon in *P. putida*.
6. **AlphaFold structural inference.** In the absence of an experimental structure, an AlphaFold model of Q88Q39 can be used to validate the predicted PLP-binding pocket, tetramer interface, and the acyl-binding β 10– β 11 loop, providing structural support ahead of wet-lab work.

Conclusion

The gene symbol **metB** is **not ambiguous** for this target: gene name, protein description, organism, family, and domain complement all consistently identify Q88Q39/PP_0659 as **cystathionine γ -synthase** of *P. putida* KT2440. The enzyme is a cytoplasmic, PLP-dependent, homotetrameric trans-sulfuration enzyme (EC 2.5.1.48; K01739) that catalyzes the **first committed step of methionine biosynthesis** — the γ -replacement of *O*-succinyl-L-homoserine with L-cysteine to give L-cystathionine plus succinate — within cysteine/methionine metabolism (pathway ppu00270, module M00017). Its catalytic PLP is anchored at Lys216, catalysis proceeds through a pyridoxamine-vinylglyoxylate intermediate that also drives a minor γ -elimination side reaction, and its physiological substrate is set by *Pseudomonas*' use of the succinyl-homoserine ester. These assignments are strongly supported by biochemistry and crystallography of close orthologs but await direct experimental confirmation for the *P. putida* protein itself.